# Unit 1 - Math Fundamentals
# 1.1 Logic

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# Outline


1. Why Logic?
2. Statements
3. Connectives and Truth Tables
4. Compound Statements
5. De Morgan's Laws
6. Converse, Inverse, and Contrapositive

# Why Logic?

# A Problem to Motivate Us

## Defining a study cohort

You are running a study and you need to say **exactly** who is eligible:

*"Patients who are diabetic AND (hypertensive OR obese), but NOT pregnant."*

This one criterion has to be:

- precise enough that two people reading it pick the **same** patients,
- and precise enough that a **computer** can pick them for you.

Ordinary English is not that precise. Logic is the language that is.

# Where This Shows Up in Data Science

## The same words, everywhere

- **Filtering data:** keeping the rows of a table that satisfy a condition.
- **Database / EHR queries:** `WHERE diabetic AND NOT pregnant`.
- **Model decisions:** a classifier ultimately answers a true/false question.
- **Proofs and definitions:** every definition in this course is a logical statement.

## What we will build

1. **Statements**: things that are true or false.
2. **Connectives**: how to combine statements (NOT, AND, OR, IF-THEN, IFF).
3. **Truth tables**: how to compute the truth value of a combination.
4. **De Morgan's laws**: how to negate a compound condition.
5. **The contrapositive**: which rearrangements of "if-then" keep its

# Statements

# What Is a Statement?

## Statement

A **statement**:

- describes a **state of affairs**,
- **alleges a fact**: it makes a claim,
- and therefore carries a **truth value**, **TRUE** or **FALSE**, never both.

## These ARE statements

- "I washed the car."
- "$x = 1$"
- "TN is a state."

## These are NOT

- "What time is it?"
- "*Venga aquí!*"
- "Blue."

**The test:** could you sensibly reply "that's true" or "that's false"?

# Writing the Two Truth Values

Every statement carries one of exactly two values. Different fields write those two values differently, the choice is one of **notation**, never of meaning.

| **Convention** | **“yes”** | **“no”** |
|---|---|---|
| written out | TRUE | FALSE |
| abbreviated | T | F |
| as bits (computing) | $1$ | $0$ |
| as spin (quantum) | up $\uparrow$ | down $\downarrow$ |

## What actually matters

There are exactly **two** values, and that is all the logic depends on. Everything we prove will hold no matter which set of labels you use.

These slides will use T and F.

# Naming Statements with Letters

Just as algebra uses $x$ to stand for a number, logic uses letters to stand for whole statements. We usually use $p$, $q$, $r$:

$$p = \text{“I washed the car”}$$
$$q = \text{“I left my house”}$$

Then instead of writing out “the statement ‘I washed the car’ is true,” we simply write

$$p = \mathsf{T}$$

**Why bother?**

Because it lets us study the structure of an argument without being distracted by what it is about. This matters because logic should always work regardless of what the statements of interest actually say.

# Two Different Equals Signs?

A fair objection, these cannot both mean the same thing:

$$p = \text{“I washed the car”}$$
$$p = \mathsf{T}$$

## They are doing different jobs

- The first **names**, “let $p$ stand for this statement.” Some books write $p :$ instead, to keep the two apart.
- The second **evaluates**, it reports the one thing about $p$ the logic uses.

## Algebra already works this way

Let $x$ count the patients in a clinic. Once you write $x = 5$, the arithmetic no longer cares what was counted, $5 + 3 = 8$ regardless.

Logic does the same, with only **two** possible values instead of infinitely many.

# Connectives and Truth Tables

# Combining Statements

## Boolean algebra

Developed by **George Boole** (1815 to 1864). Ordinary algebra combines *numbers* with $+$ and $\times$; Boolean algebra combines *statements* with NOT, AND, and OR.

The combining symbols are called **connectives**:

| **Symbol** | **Name** | **Read as** |
|---|---|---|
| $\neg p$ | negation | "not $p$" |
| $p \wedge q$ | conjunction | "$p$ and $q$" |
| $p \vee q$ | disjunction | "$p$ or $q$" |
| $p \rightarrow q$ | implication | "if $p$, then $q$" |
| $p \leftrightarrow q$ | bi-implication | "$p$ if and only if $q$" |

# The Truth Table

## Truth table

A **truth table** lists **every** possible combination of truth values for the inputs, and gives the resulting truth value of the output. It is the complete definition of a connective, nothing is left to interpretation.

## How to read one

Each **row** is one possible assignment of truth values to the inputs. Each **column** is one statement, an input, an intermediate piece, or the final result.

Every connective on the previous slide is defined by exactly one such table, and we build them next.

# Negation (NOT)

The **negation** of $p$ is written $\neg p$ (or NOT $p$). It flips the truth value.

## Example

$$p = \text{“I went to the store”}$$
$$\neg p = \text{“I did NOT go to the store”}$$

| $p$ | $\neg p$ |
|---|---|
| T | F |
| F | T |

Negating twice returns you to where you started:

$$\neg(\neg p) = p$$

# Conjunction (AND)

The **conjunction** of $p$ and $q$ is written $p \wedge q$, read "$p$ and $q$."

## Example

$$p = \text{"I went to the store"}$$
$$q = \text{"I left my house"}$$
$$p \wedge q = \text{"I went to the store AND I left my house"}$$

| $p$ | $q$ | $p \wedge q$ |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | F |
| F | F | F |

## Note

$p \wedge q$ is TRUE only when **both** inputs are TRUE.

# Disjunction (OR)

The **disjunction** of $p$ and $q$ is written $p \vee q$, read "$p$ or $q$."

## Example

$$p = \text{"I went to the store"}$$
$$q = \text{"I left my house"}$$
$$p \vee q = \text{"I went to the store OR I left my house"}$$

| $p$ | $q$ | $p \vee q$ |
|---|---|---|
| T | T | T |
| T | F | T |
| F | T | T |
| F | F | F |

## Note

$p \vee q$ is FALSE only when **both** inputs are FALSE.

# Inclusive vs. Exclusive OR

Look again at the first row of the OR table: when **both** are true, $p \vee q$ is TRUE. That is a deliberate choice, and it is not how we always speak.

## Two different "or"s

- **Inclusive OR** ($p \vee q$): true if *at least* one of $p, q$ is true.
  *"Applicants with a nursing degree or clinical experience"*, having both is fine.
- **Exclusive OR** (XOR): true if *exactly* one of $p, q$ is true.
  *"Soup or salad"*, pick one, not both.

| $p$ | $q$ | $p \vee q$ (inclusive) | $p$ XOR $q$ (exclusive) |
|---|---|---|---|
| T | T | T | F |
| T | F | T | T |
| F | T | T | T |
| F | F | F | F |

**In mathematics, "or" always means the inclusive one** unless we say

# Implication (IF-THEN)

Implication is written $p \to q$, read "if $p$, then $q$" or "$p$ implies $q$."

- $p$ is the **antecedent** (the hypothesis).
- $q$ is the **consequent** (the conclusion).

## Example

$$p = \text{"I went to the store"}$$
$$q = \text{"I left my house"}$$
$$p \to q = \text{"IF I went to the store, THEN I left my house"}$$

Think of this as a **promise**. The only way to break the promise is for $p$ to happen and $q$ to fail.

# Implication: The Truth Table

| $p$ | $q$ | $p \to q$ |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | T |
| F | F | T |

**Note**

$p \to q$ is FALSE only when the antecedent is TRUE and the consequent is FALSE, i.e. only when it is a **lie**.

**The surprising rows**

If $p$ is FALSE, the implication is TRUE no matter what, the promise was never tested.

*"If I went to the store, then I left my house."* If I never went to the store, I have not broken my word. This is called being **vacuously true**.

# Bi-implication (IF AND ONLY IF)

Bi-implication is written $p \leftrightarrow q$, read "$p$ if and only if $q$" (often "$p$ iff $q$"). The implication runs **both ways**:

$$p \leftrightarrow q \quad = \quad (p \rightarrow q) \wedge (q \rightarrow p)$$

| $p$ | $q$ | $p \rightarrow q$ | $q \rightarrow p$ | $p \leftrightarrow q$ |
|---|---|---|---|---|
| T | T | T | T | T |
| T | F | F | T | F |
| F | T | T | F | F |
| F | F | T | T | T |

**Note**

The last column is the **AND** of the two middle ones, true only where *both* implications hold. So $p \leftrightarrow q$ is TRUE exactly when $p$ and $q$ **match**.

This is the connective behind every mathematical **definition**, "even *if and only if* divisible by 2."

# How Many Inputs? Unary and Binary

That is every connective we need. Looking back over them, notice **how many statements each one takes**, a property called its **arity**.

## Unary vs. binary

- A **unary** operator takes **one** input. Only $\neg$ is unary.
- A **binary** operator takes **two** inputs, $\wedge, \vee, \rightarrow, \leftrightarrow$, and XOR are all binary.

## The same split in arithmetic

- The negative sign in $-x$ is **unary**, one number in, one out.
- $+$, $-$, $\times$, $\div$ are **binary**, two numbers in, one out.

This will matter shortly. If $\wedge$ only ever joins **two** statements, what could $p \wedge q \wedge r$ possibly mean?

# Summary of the Connectives

| $p$ | $q$ | $\neg p$ | $p \wedge q$ | $p \vee q$ | $p \rightarrow q$ | $p \leftrightarrow q$ | XOR |
|---|---|---|---|---|---|---|---|
| T | T | F | T | T | T | T | F |
| T | F | F | F | T | F | F | T |
| F | T | T | F | T | T | F | T |
| F | F | T | F | F | T | T | F |

## Worth memorizing as one-liners

- AND: true only when **both** are true.
- OR: false only when **both** are false.
- IF-THEN: false only when **true** implies **false**.
- IFF: true only when they **match**. XOR: true only when they **differ**.

# Compound Statements

# Building Bigger Expressions

Connectives can be nested, exactly like arithmetic. Parentheses decide the order:

$$\neg(p \wedge q) \qquad \text{is not the same as} \qquad (\neg p) \wedge q$$

## Strategy for a compound truth table

1. Write one column for each input statement, and give every combination of their truth values its own row.
2. Add a column for each **intermediate** piece, working from the inside out.
3. The last column is the answer.

Doing this by hand a few times is how the rules stop feeling arbitrary. After that, let a computer do it, see the notebook `U1-1_Logic-1_ConnectivesAndTruthTables.ipynb`.

# Chaining ANDs and ORs

$\wedge$ is binary, so $p \wedge q \wedge r$ must be read as *two* ANDs. Both $\wedge$ and $\vee$ are well behaved in two ways that make this harmless.

## Commutative, *order* does not matter

$$p \wedge q \;=\; q \wedge p \qquad\qquad p \vee q \;=\; q \vee p$$

Exactly as $a + b = b + a$ and $a \times b = b \times a$.

## Associative, *grouping* does not matter

$$(p \wedge q) \wedge r \;=\; p \wedge (q \wedge r)$$

Exactly as $(1 + 2) + 3 = 1 + (2 + 3)$, which is why we write $1 + 2 + 3$.

Together these let us drop the parentheses entirely: $p \wedge q \wedge r \wedge s$ says "all four are true" and $p \vee q \vee r \vee s$ says "at least one of the four is," in whatever order you like.

# Where the Analogy Stops

Implication fails **both** tests, exactly as subtraction does.

## Order matters

$$p \to q \;\neq\; q \to p \qquad\qquad 8 - 3 \;\neq\; 3 - 8$$

Swapping the two sides gives the **converse**, a different statement, as we will see in a later section.

## Grouping matters

$$(p \to q) \to r \;\neq\; p \to (q \to r)$$

Exactly as $(8 - 3) - 2 = 3$ but $8 - (3 - 2) = 7$.

## So

**Never write** $p \to q \to r$**.** Mixing $\wedge$ with $\vee$ needs parentheses too, as the next two slides show.

# Why Parentheses Matter

Are $(p \wedge q) \vee r$ and $p \wedge (q \vee r)$ really different? Three statements means eight combinations, one row each. Build both columns and compare.

| $p$ | $q$ | $r$ | $p \wedge q$ | $(p \wedge q) \vee r$ | $q \vee r$ | $p \wedge (q \vee r)$ |
|---|---|---|---|---|---|---|
| T | T | T | T | T | T | T |
| T | T | F | T | T | T | T |
| T | F | T | F | T | T | T |
| T | F | F | F | F | F | F |
| F | T | T | F | T | T | F |
| F | T | F | F | F | T | F |
| F | F | T | F | T | T | F |
| F | F | F | F | F | F | F |

**The last two columns disagree on two rows**

Both are rows where $p$ is FALSE and $r$ is TRUE. The parentheses decide whether $r$ is allowed to carry the whole statement **by itself**, or must still get past $p$ first.

# What That Difference Means

Read the two groupings aloud, with $d$ = diabetic, $h$ = hypertensive, $o$ = obese:

**$(d \wedge h) \vee o$**

"Diabetic **and** hypertensive, **or else** obese."

Being obese is enough on its own. The other two never have to hold.

**$d \wedge (h \vee o)$**

"Diabetic, **and** on top of that hypertensive or obese."

Diabetes is mandatory. Being obese only helps once you already have it.

**One patient tells them apart**

A patient who is **obese but not diabetic** satisfies the first and fails the second, exactly the $p$ FALSE, $r$ TRUE rows from the table.

Same three conditions, same three words, one pair of parentheses, and a different cohort.

# Worked Example

Build the truth table for $\neg p \vee q$.

| $p$ | $q$ | $\neg p$ | $\neg p \vee q$ |
|---|---|---|---|
| T | T | F | T |
| T | F | F | F |
| F | T | T | T |
| F | F | T | T |

## Compare this with the implication table

The final column is T, F, T, T, **identical** to $p \to q$. So

$$p \to q \;=\; \neg p \vee q$$

Implication is not really a new idea; it can be built from NOT and OR.

# Logical Equivalence

## Logical equivalence

Two compound statements are **logically equivalent** if their truth tables have the **same final column**, they agree on every possible input.

Equivalent statements can be swapped for one another freely. That is what makes them useful: we can **rewrite** a complicated condition as a simpler one without changing its meaning.

## This is a proof technique

To prove two expressions are equivalent, **compute both truth tables and compare**. With finitely many rows, checking them all is a complete proof.

# Exercise

## Try it

Build the truth table for

$$(p \wedge q) \vee (\neg p \wedge \neg q)$$

Then answer: which connective from earlier is this equivalent to?

| $p$ | $q$ | $p \wedge q$ | $\neg p$ | $\neg q$ | $\neg p \wedge \neg q$ | full expression |
|---|---|---|---|---|---|---|
| T | T | | | | | |
| T | F | | | | | |
| F | T | | | | | |
| F | F | | | | | |

# Exercise: Solution

| $p$ | $q$ | $p \wedge q$ | $\neg p$ | $\neg q$ | $\neg p \wedge \neg q$ | full expression |
|---|---|---|---|---|---|---|
| T | T | T | F | F | F | T |
| T | F | F | F | T | F | F |
| F | T | F | T | F | F | F |
| F | F | F | T | T | T | T |

The last column is the OR of the $p \wedge q$ and $\neg p \wedge \neg q$ columns: a row is TRUE when either of those is true.

## Answer

The final column is T, F, F, T, the table for **bi-implication**:

$$(p \wedge q) \vee (\neg p \wedge \neg q) \;=\; p \leftrightarrow q$$

Which makes sense: the expression says "both true, or both false", that is exactly "$p$ and $q$ match."

# De Morgan's Laws

# Negating AND and OR

How do you negate a compound condition? The naive guess, "flip everything", is wrong in an interesting way.

## De Morgan's Laws

$$\neg(p \wedge q) = \neg p \vee \neg q$$
$$\neg(p \vee q) = \neg p \wedge \neg q$$

## In words

Push the negation inward, and **the connective flips**: AND becomes OR, OR becomes AND.

*"It is not the case that I am both hungry and tired" = "I am not hungry, or I am not tired."*

# Proving De Morgan's First Law

Claim: $\neg(p \wedge q) = \neg p \vee \neg q$. Build both truth tables and compare.

| $p$ | $q$ | $p \wedge q$ | $\neg(p \wedge q)$ | $\neg p$ | $\neg q$ | $\neg p \vee \neg q$ |
|---|---|---|---|---|---|---|
| T | T | T | F | F | F | F |
| T | F | F | T | F | T | T |
| F | T | F | T | T | F | T |
| F | F | F | T | T | T | T |

The 4th and 7th columns match on every row, so the two statements are logically equivalent. ■

The second law is proved the same way, a good exercise.

# Why We Care

## Negating a cohort definition

Eligible: *diabetic* $\wedge$ *hypertensive*

Who is **ineligible**? By De Morgan:

$$\neg(\text{diabetic} \wedge \text{hypertensive}) = \neg\text{diabetic} \ \vee \ \neg\text{hypertensive}$$

Not "patients who are neither", patients who are **missing either one**. A very easy mistake to make in a query, and an expensive one.

# Exercise

## Try it

Negate the full cohort criterion:

$$\text{diabetic} \wedge (\text{hypertensive} \vee \text{obese}) \wedge \neg\text{pregnant}$$

Push the negation all the way in, so that no $\neg$ is left sitting outside a parenthesis.

# Exercise: Solution

Push the negation in one layer at a time. The outer $\wedge$ chain flips to a $\vee$ chain:

$$\begin{aligned}
&\neg\big(d \wedge (h \vee o) \wedge \neg g\big) \\
&\quad = \ \neg d \ \vee \ \neg(h \vee o) \ \vee \ \neg(\neg g) && \text{(De Morgan on the ANDs)} \\
&\quad = \ \neg d \ \vee \ (\neg h \wedge \neg o) \ \vee \ g && \text{(De Morgan again, and } \neg\neg g = g\text{)}
\end{aligned}$$

**In words**

A patient is **ineligible** if they are not diabetic, **or** they are neither hypertensive nor obese, **or** they are pregnant.

Note the inner $\vee$ flipped to an $\wedge$ while the outer $\wedge$s flipped to $\vee$s, every connective changes, at every level.

# Converse, Inverse, and Contrapositive

# Three Relatives of an Implication

Start with an implication and rearrange it three ways:

| **Statement** | $p \to q$ |
|---|---|
| **Converse** | $q \to p$ |
| **Inverse** | $\neg p \to \neg q$ |
| **Contrapositive** | $\neg q \to \neg p$ |

## Example

- **Statement:** If it is raining, then the ground is wet.
- **Converse:** If the ground is wet, then it is raining. *(not the same!)*
- **Inverse:** If it is not raining, then the ground is not wet. *(not the same!)*
- **Contrapositive:** If the ground is not wet, then it is not raining. *(same!)*

# Only the Contrapositive Is Equivalent

| $p$ | $q$ | $p \to q$ *statement* | $q \to p$ *converse* | $\neg p \to \neg q$ *inverse* | $\neg q \to \neg p$ *contrapositive* |
|---|---|---|---|---|---|
| T | T | T | T | T | T |
| T | F | F | T | T | F |
| F | T | T | F | F | T |
| F | F | T | T | T | T |

## Read off the table

- Statement and **contrapositive** match, always equivalent.
- Converse and **inverse** match each other, but neither matches the original.

Confusing a statement with its converse is one of the most common reasoning errors there is, "the test was positive, so the patient has the disease" is exactly this mistake.

# Review

## What we covered

1. A **statement** carries one of exactly two truth values.
2. **Connectives** $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$ combine them; a **truth table** defines each completely.
3. $\neg$ is **unary**, the rest **binary**; $\wedge$ and $\vee$ are commutative and associative, $\rightarrow$ is neither.
4. Statements are **logically equivalent** when their truth tables agree, which is also how we prove it.
5. **De Morgan's laws** negate AND/OR by flipping the connective.
6. Only the **contrapositive** is equivalent to the original implication.

**Notebook:** `U1-1_Logic-1_ConnectivesAndTruthTables.ipynb`
**Next:** Unit 1-2, **Sets**, the same logic applied to collections, where our cohort finally becomes an actual set.